

Realistic Evaluation of Coupling Loss Between Different Optical Fibers

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Summary

Coupling between two multimode optical fibers with different numerical aperture and core diameter and including lateral displacement is studied. Calculations are performed for fibers with any index profile and any power distribution among ray congruences. The propagation effects in the second fiber are also taken into account in a rigorous way. Finally asymptotic analytic expressions of losses for small values of mismatch and error parameters are given together with a discussion on the minimization of coupling losses.

1 Introduction

The problem of fiber-fiber jointing is of great importance in optical fiber telecommunications since coupling losses are comparable with attenuation of one kilometer of most recent fibers. Thus a realistic theoretical analysis of the problem seems necessary.

The most critical parameters, which will be considered in detail in this paper, are mismatch in core diameter, mismatch in maximum numerical aperture and lateral displacement. For what concerns these error parameters, analyses published in the literature are at present insufficient. In fact in many cases [1, 2] the hypothesis of uniform power distribution among ray congruences (or modes) is assumed. This constitutes a first approach to the problem which is not realistic in general, thus providing too pessimistic predictions. On the other hand, analyses which take into account non-uniform power distributions either are not very clear [3, 4], with results which sometimes represent only a rough approximation to the correct ones or are empirical models [5], which neglect propagation effects in the second fiber. Moreover, while it is important to obtain coupling loss due to the splice or the joint itself (that is, considering the total amount of power accepted and guided by the second fiber), one has to evaluate the whole loss caused by coupling (considering also the additional attenuation produced by the splice or joint in the second fiber). This last coupling loss is here correctly derived and results for both kinds of losses are provided.

The present paper considers these aspects taking into account fibers with any index profile and any distribution of power among ray congruences. A preliminary discussion on radiance distribution precedes the general study of coupling loss. Finally analytical expressions are given for small values of error parameters together with a discussion about the problem of minimizing coupling loss.

2 Radiance distribution inside the fiber

It is known [6] that every ray congruence inside a fiber can be determined by means of the following two parameters

$$h \equiv n(r')r' \sin \vartheta \sin \varphi \quad \text{and} \quad k \equiv n(r') \cos \vartheta \quad (1)$$

having introduced a system of cylindrical polar coordinates (r', ψ, z) coaxial to the fiber and the angular polar co-ordinates (ϑ, φ) which characterize the direction of each ray (ϑ is the angle of the ray with respect to the fiber axis and φ the azimuthal angle of the ray with respect to the radius vector). $n(r')$ represents the index profile of the fiber core.

In general the power distribution $q(h, k)$ among the ray congruences depends on both h and k and can be deduced from the radiance distribution $R(r', \psi, \vartheta', \varphi)$ (ϑ' is the corresponding value of ϑ in air) of the fiber through

$$q(h, k) = 4k \int_{r_0}^{r_1} \frac{dr'}{\sqrt{n^2(r') - k^2 - h^2/r'^2}} \int_0^{2\pi} d\psi R \left(r', \psi, \arcsin \sqrt{n^2(r') - k^2}, \arcsin \frac{h}{r' \sqrt{n^2(r') - k^2}} \right), \quad (2)$$

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r'_0 and r'_1 being the radii of the two caustics of the guided congruence.

The total power inside the fiber is given by

$$W = \int_0^a r' dr' \int_0^{2\pi} d\psi \int_0^{2\pi} d\varphi \int_0^{\sqrt{n^2(r') - n_1^2}} \sin \vartheta' \cos \vartheta' d\vartheta' R(r', \psi, \vartheta', \varphi) = \int_{n_1}^{n_0} dk \int_0^{h_e(k)} dh Q(h, k) \quad (3)$$

where a is the core radius; n_1 the refractive index of cladding, $h_e(k)$ represents for every k the h -value for circular helicoidal rays. In (3) as well as in the following leaky skew rays are not considered.

From (2) we have that when R does not depend on φ (and thus on h) a radiance distribution against k can be explicitly obtained. In performing this it seems more convenient to introduce the following congruence variable (instead of k)

$$x \equiv \frac{n_0^2 - k^2}{\Delta^2} \quad (4)$$

$\Delta \equiv \sqrt{n_0^2 - n_1^2}$ is the maximum numerical aperture; x ranges from 0 (axial rays) to 1 (rays at the guidance limit condition). Thus the following radiance distribution as a function of x can be obtained (integrating (2) over h)

$$R(x) = \frac{2}{\pi i^2(x)} \int_0^\pi d\psi \int_0^{i(x)} R(r, \psi, x) r dr, \quad (5)$$

where

$$r \equiv \frac{r'}{a}, \quad (6)$$

$$R(r, \psi, x) \equiv R(r', \psi, \arcsin \sqrt{\Delta^2 x - [n_0^2 - n^2(r')]}), \quad (7)$$

and $i(x)$ is obtained from $r'_M(n)$, the inverse function of the index profile $n(r')$, through

$$i(x) = r'_M(\sqrt{n_0^2 - \Delta^2 x})/a. \quad (8)$$

Then the total power guided by the fiber is given by

$$W = \pi^2 a^2 \Delta^2 \int_0^1 R(x) i^2(x) dx. \quad (9)$$

The index profile can be written as

$$n^2(r') \equiv n_0^2 - \Delta^2 f(r) \quad (10)$$

where generally $f(r)$ is an increasing function such that: $f(0) = 0$ and $f(1) = 1$; thus $i(x)$ is the inverse function of $f(r)$.

The assumption that R in (2) is independent of h will be maintained in the following and in practice can be fulfilled in many cases of interest. In any case the present analysis represents a first approach to the problem.

$R(x)$ is assumed known in the calculation of coupling losses. It can be obtained both analytically and experimentally. $R(x)$ can be considered as constant for short, uniformly excited fibers (see (5)–(7)). For long lengths of fibers the steady-state power distribution $R_0(x)$ can be considered as a good approximation of $R(x)$. For intermediate fiber lengths z the actual expression of $R(x)$ can be deduced from an analysis of the scattering phenomenon [7] and can be expressed by a superposition of eigenfunctions $R_n(x)$ (β_n being the relative eigenvalues)

$$R(x, z) = \sum_{n=0}^{\infty} A_n e^{-\beta_n z} R_n(x), \quad (11)$$

where the amplitudes A_n are deduced from the initial distribution $R(x, 0) = R(x)$ through

$$A_n = \frac{\int_0^1 R(x) R_n(x) i^2(x) dx}{\int_0^1 R_n^2(x) i^2(x) dx}. \quad (12)$$

Note that (12) represents the generalization to any profile of a similar equation given in [7], which however is rigorous only for parabolic profiles.

$R(x)$ can also be obtained experimentally through near-field intensity measurements (as proposed in [8] but for power law profiles only), after that the index profile is known. It is known [6] that in general the radial distribution of power is given by

$$w(r') = \int_{n_1}^{n(r')} dk \int_0^{r' \sqrt{n^2(r') - k^2}} dh \frac{Q(h, k)}{\sqrt{n^2(r') - k^2 - h^2/r'^2}} \left\{ \int_{r'_0}^{r'_1} \frac{dr''}{\sqrt{n^2(r'') - k^2 - h^2/r''^2}} \right\}^{-1}. \quad (13)$$

If a degeneracy on h exists, we have

$$Q(h, k) = 8\pi k R \left(\frac{n_0^2 - k^2}{\Delta^2} \right) \int_{r'_0}^{r'_1} \frac{dr''}{\sqrt{n^2(r'') - k^2 - h^2/r''^2}} \quad (14)$$

thus (13) becomes

$$w(r') = 4\pi^2 r' \int_{n_1}^{n(r')} R \left(\frac{n_0^2 - k^2}{\Delta^2} \right) k dk \quad (15)$$

and the near-field intensity is

$$I(r) = \frac{w(r')}{2\pi r'} = \pi \Delta^2 \int_{f(r)}^1 R(x) dx. \quad (16)$$

$R(x)$ can be deduced from (16)

$$R(x) = -\pi \Delta^2 i'(x) \left(\frac{dI}{dr} \right)_{r=f(x)} \quad (17)$$

(the prime indicates the derivative with respect to the argument). Calling $I_0(r)$ the near-field intensity for uniform excitation ($R(x) = R_0$), the actual distribution $R(x)$ can be obtained by comparing the two near-field measurements I and I_0 in the following way

$$R(f(r)) = R_0 \frac{dI/dr}{dI_0/dr} \quad (18)$$

provided that I_0 reproduces the index profile ($I_0 = \pi \Delta^2 (1 - f(r))$).

3 Fiber-fiber coupling loss

The study of fiber-fiber coupling loss will be performed for fibers with the same (generic) index profile $f(r)$, but with possible different values of core radius a , numerical aperture Δ considering lateral displacement (which is most critical compared to angular misalignment and longitudinal separation [2, 9]). A bar will indicate the quantity pertaining to the second (receiving) fiber.

Let us call W_A the optical power launched by the first fiber and $\bar{W}_A$ the power received and guided by the second fiber

$$W_A = \pi^2 a^2 \Delta^2 \int_0^1 R(x) i^2(x) dx, \quad (19)$$

$$\bar{W}_A = \pi^2 \bar{a}^2 \bar{\Delta}^2 \int_0^1 \bar{R}(x) i^2(x) dx,$$

where $R(x)$ is the radiance distribution at the output end of the first fiber and $\bar{R}(x)$ the corresponding radiance distribution at the input end of the second fiber, related to $R(x)$ through (from (5))

$$\bar{R}(\bar{x}) = \frac{2}{\pi i^2(\bar{x})} \int_0^\pi d\bar{\Psi} \int_0^{\bar{x}} R(x(\bar{r}, \bar{\Psi}, \bar{x})) \bar{r} d\bar{r}; \quad (20)$$

$x(\bar{r}, \bar{\Psi}, \bar{x})$ represents the parameter x in the first fiber corresponding to the parameter $\bar{x}$ in the second fiber for a ray in the point $(\bar{r}, \bar{\Psi})$ and is given by (after (4))

$$x(\bar{r}, \bar{\Psi}, \bar{x}) = (\bar{\Delta}/\Delta)^2 (\bar{x} - f(\bar{r})) + f(r(\bar{r}, \bar{\Psi})); \quad (21)$$

where the value of r corresponding to $(\bar{r}, \bar{\Psi})$ can be obtained by means of simple geometrical considerations, taking into account the lateral displacement u (normalized to a)

$$r(\bar{r}, \bar{\Psi}) = \sqrt{\left(\frac{\bar{r}\bar{a}}{a}\right)^2 + u^2 - 2\bar{r}\bar{u}\frac{\bar{a}}{a} \cos \bar{\Psi}}. \quad (22)$$

The loss L_A due to the joint only and without considering the propagation effects in the second fiber, is hence given by

$$L_A \equiv \frac{W_A - \bar{W}_A}{W_A} = 1 - \left(\frac{\bar{a}\bar{\Delta}}{a\Delta}\right)^2 \frac{\int_0^1 \bar{R} i^2 dx}{\int_0^1 R i^2 dx} \quad (23)$$

The joint causes also an additional attenuation in the second fiber that, if this fiber is long enough to reach the steady state, must be taken into account. L_B is the loss of the joint including such an additional attenuation. Let us call W_B the total optical power at the output of the second fiber in the case in which neither mismatch nor coupling errors are present; we have (z is counted starting from the joint)

$$W_B(z) = \pi^2 a^2 \Delta^2 \int_0^1 R(x, z) i^2(x) dx \quad (24)$$

where $R(x, z)$ is given by (11) with A_n given by (12). Calling $\bar{W}_B$ the total optical power present at the output of the second fiber, we have

$$\bar{W}_B(z) = \pi^2 \bar{a}^2 \bar{\Delta}^2 \int_0^1 \bar{R}(x, z) i^2(x) dx \quad (25)$$

where $\bar{R}(x, z)$ is given by an equation similar to (11) in which coefficients $\bar{A}_n$ can be obtained from (12) considering that now $R(x)$ must be replaced by $\bar{R}(x)$ of (20). L_B is given by

$$L_B \equiv \frac{W_B - \bar{W}_B}{W_B} \quad (26)$$

and for large values of z (when only the first term in the series (11) has to be retained) we have

$$L_B = 1 - \left(\frac{\bar{a}\bar{\Delta}}{a\Delta}\right)^2 \frac{\bar{A}_0}{A_0} \quad (27)$$

$$= 1 - \left(\frac{\bar{a}\bar{\Delta}}{a\Delta}\right)^2 \frac{\int_0^1 \bar{R} R_0 i^2 dx}{\int_0^1 R R_0 i^2 dx}.$$

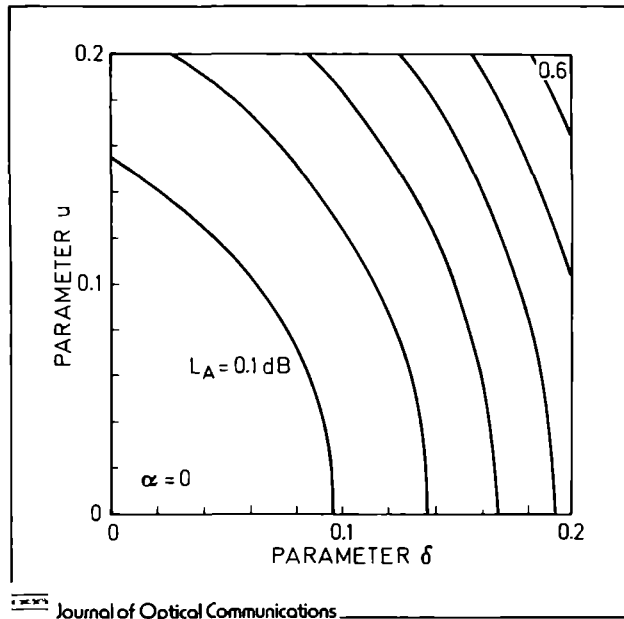
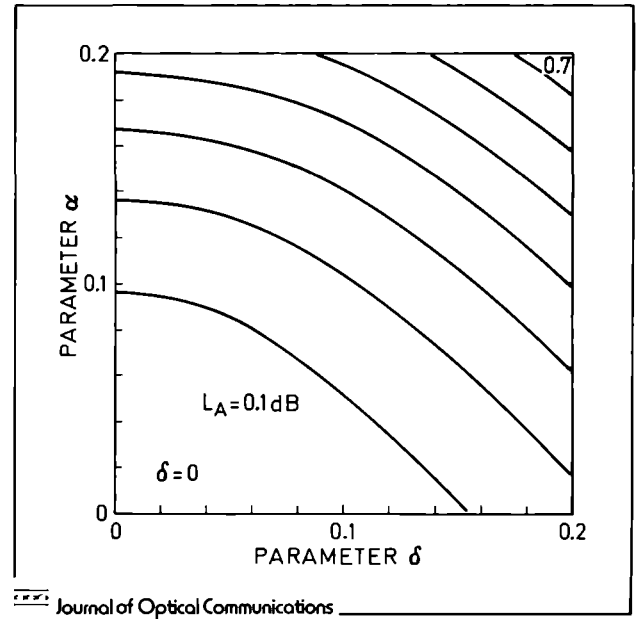
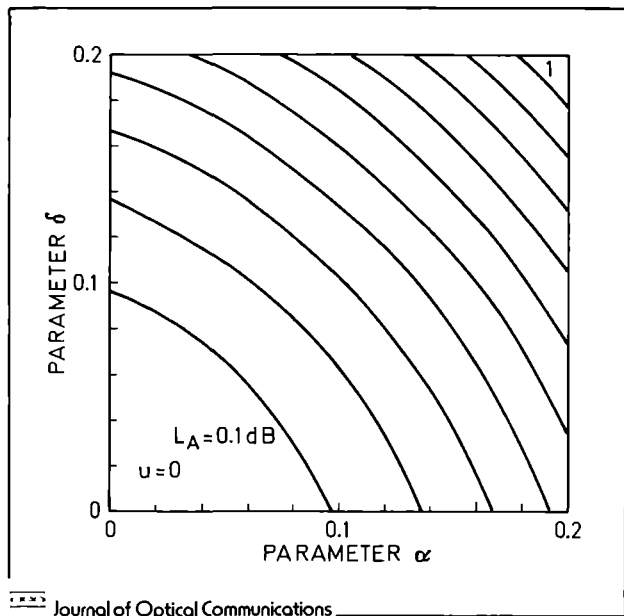


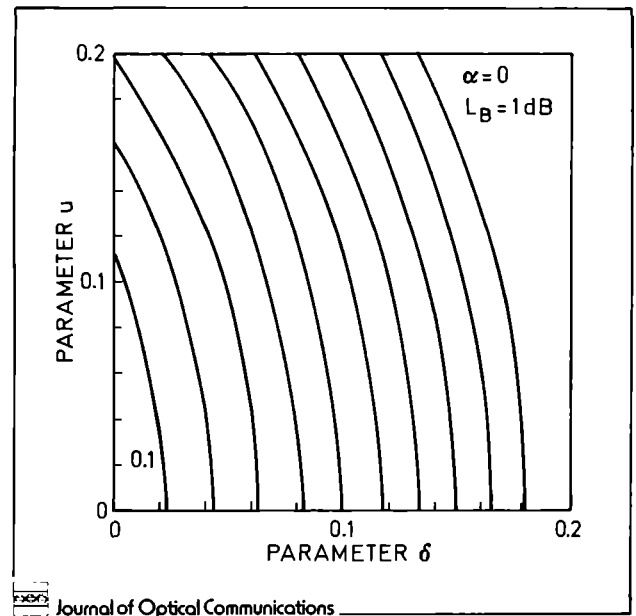
Fig. 1: Plane at constant coupling joint loss L_A in the $u-\delta$ plane, for $\alpha=0$. The curves, starting from 0.1 dB, are separated by a 0.1 dB step



The same as Fig. 1, but in the $\alpha-u$ plane for $\delta=0$



The same as Fig. 1, but in the $\delta-\alpha$ plane for $u=0$



The same as Fig. 1, but for the overall loss L_B

This expression of L_B considers the whole loss due to the joint when the steady-state in the second fiber is reached; that is the maximum loss that can be produced by the joint.

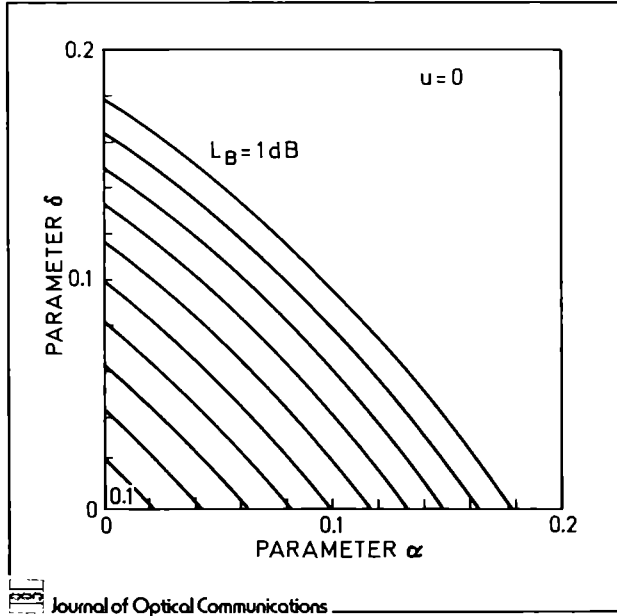
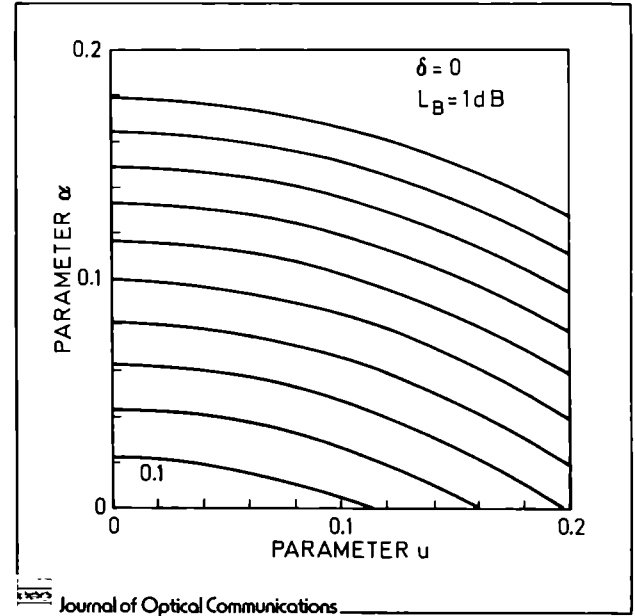
L_A can be considered as a good estimate of coupling loss when the second fiber is short with respect to steady-state length; while L_B is more realistic for long lengths of fiber. L_A and L_B will be calculated as functions of the normalized lateral displacement u and the relative mismatches α and δ according to a and Δ , respectively

$$\alpha \equiv \frac{a - \bar{a}}{a}, \quad \delta \equiv \frac{\Delta - \bar{\Delta}}{\Delta}. \quad (28)$$

Using the steady-state power distribution of [7], which seems realistic, at least for parabolic profiles,

$$R(x) = R_0(x) = \frac{2 R_0(0)}{j_{11} \sqrt{x}} J_1(j_{11} \sqrt{x}) \quad (29)$$

(J_q is the Bessel function of the first kind and q -th order and j_{q1} is its first zero) Figs. 1 to 6 can be obtained showing curves at constant coupling loss (joint loss L_A for Figs. 1 to 3, and overall loss L_B for Figs. 4 to 6) in the coupling error planes $u-\alpha$ (Figs. 1 and 4, $\delta=0$); $\alpha-\delta$ (Figs. 2 and 5, $u=0$); $\delta-u$ (Figs. 3 and 6, $\alpha=0$). u is the less critical among the error parameters. One should

Fig. 5: The same as Fig. 4, but in the δ - α plane for $u=0$ Fig. 6: The same as Fig. 4, but in the α - u plane for $\delta=0$

notice that α and δ can be interchanged when $u=0$. Figures 2 and 5 are symmetric with respect to the bisecting line of the axis α and δ .

4 Small values of coupling errors

The case in which α , δ and u are very small (with respect to unity) is important, since it is often met in practice. In this section the simultaneous presence of α , δ and u will be considered; afterwards a particular mention will be devoted to the common case of coupling between equal fibers ($\alpha=\delta=0$) in which only lateral displacement will be taken into account.

4.1 General case

Analytical expressions of L_A and L_B can be derived when small values of α , δ and u are considered. Expanding (21)–(23) and (27) up to the second order we have

$$\begin{aligned} x(\bar{r}, \bar{\psi}, \bar{x}) \simeq & \bar{x} - (2\delta - \delta^2) [\bar{x} - f(\bar{r})] \\ & - \left(\alpha \bar{r} + u \cos \bar{\psi} - \frac{u^2}{2\bar{r}} \sin^2 \bar{\psi} \right) \cdot f'(\bar{r}) \\ & + (\alpha \bar{r} + u \cos \bar{\psi})^2 f''(\bar{r})/2 \end{aligned} \quad (30)$$

$$\begin{aligned} L_{A,B} = & 1 - (1 - 2\alpha - 2\delta + 4\alpha\delta + \alpha^2 + \delta^2) \cdot \\ & \cdot (1 - N_{A,B}/D_{A,B}) \end{aligned} \quad (31)$$

where

$$D_A = \int_0^1 R i^2 dx; \quad D_B = \int_0^1 R R_0 i^2 dx \quad (32)$$

and (after (20))

$$N_A = \frac{2}{\pi} \int_0^1 d\bar{x} \int_0^\pi d\bar{\psi} \int_0^{i(\bar{x})} \{R(\bar{x}) - R(x(\bar{r}, \bar{\psi}, \bar{x}))\} \bar{r} d\bar{r}, \quad (33)$$

$$N_B = \frac{2}{\pi} \int_0^1 d\bar{x} \int_0^\pi d\bar{\psi} \int_0^{i(\bar{x})} \{R(\bar{x}) - R(x(\bar{r}, \bar{\psi}, \bar{x}))\} R_0(\bar{x}) \bar{r} d\bar{r}. \quad (34)$$

The integrand in (33) and (34) can be expanded according to (30) up to the second order terms in α , δ and u , assuming that $R(1)=0$ and $R_0(1)=0$. Caution must be used in expanding N_A for $\alpha \leq u$ because, owing to the fact that the derivative of $R(x)$ is not continuous for $x=1$ ($R'(1) \neq 0$ and $R(x)=0$ for $x>1$), a negative term is illegally added to N_A .

The corrected results for L_A are for $\alpha \geq u$

$$\begin{aligned} L_A = & \left[\alpha^2 \int_0^1 f^2(r) r^3 dr + 4\delta(\alpha + \delta) \int_0^1 (1 - f(r))^2 r dr \right. \\ & \left. + \frac{u^2}{2} \int_0^1 f^2(r) r dr \right] \frac{|R'(1)|}{\int_0^1 R i^2 dx}; \end{aligned} \quad (35)$$

and (for $\alpha, \delta \ll u$)

$$L_A = \left[\frac{u^2}{4} \int_0^1 f^2(r) r dr \right] \frac{|R'(1)|}{\int_0^1 R i^2 dx}; \quad (36)$$

(35) and (36) exhibit a quadratic behaviour of L_A for small values of the error parameters; this is strictly due to the fact that $R(1)=0$.

A particular mention must be made to the mismatch combination $\alpha - \delta$ in the case of parabolic profile; in which (35) becomes

$$L_A = \left[\frac{2}{3} (\alpha^2 + \delta^2 + \alpha\delta) + \frac{u^2}{2} \right] \frac{|R'(1)|}{\int_0^1 R i^2 dx}, \quad (\alpha \geq u), \quad (37)$$

which shows a symmetry in the interchange of α and δ . Note that in (37) α and δ combine each other in an intermediate way between $(\alpha + \delta)^2$ and $(\alpha^2 + \delta^2)$. Results of (35) for parabolic profiles are in agreement with [4], but this agreement ceases in the case of (37) (for α and δ only) in which our analysis seems more rigorous. For what concerns L_B compact expressions have been obtained in the following two cases: for $\alpha = \delta = 0$

$$L_B = \frac{u^2}{2} \frac{\left\{ \int_0^1 dx R'(x) R'_0(x) \int_0^{i(x)} f^2(r) r dr \right\}}{\int_0^1 R(x) R_0(x) i^2(x) dx} \quad (38)$$

and for $R(x) = R_0(x)$

$$L_B = \alpha + \delta + \frac{u^2}{2} \frac{\left\{ \int_0^1 dx R'^2(x) \int_0^{i(x)} f^2(r) r dr \right\}}{\int_0^1 R_0^2(x) i^2(x) dx} \quad (39)$$

L_B exhibits a quadratic behaviour in u but a linear one in α and δ which is independent of the index profile and the radiance distribution (if $R_0(1) = 0$). L_B is always greater than corresponding L_A but it is smaller than the value of L_A considering uniform radiance distributions.

4.2 Lateral displacement

The case of coupling between two similar fibers is common in practice. As mentioned, the lateral displacement is the most critical coupling error. The behaviour of coupling for small values of u depends strongly on $R(x)$. Asymptotic expressions of L_A and L_B when $R(1) = 0$ have been given in (36) and (38). Besides when $R(1) \neq 0$ the following expression of L_A can be derived

$$L_A = \frac{2u}{\pi} R(1) \frac{1 - \int_0^1 f(r) dr}{\int_0^1 R(x) i^2(x) dx}; \quad (40)$$

while the expression of L_B when $R(x)$ is constant equates (36) of L_A with $R(x)$ replaced by $R_0(x)$. That is for $R(1) \neq 0$ L_B retains the quadratic behaviour in u , while L_A exhibits a linear one.

A case often met in practice is the jointing of two long fibers; then (38) with $R(x) = R_0(x)$ gives a good estimate of coupling loss for small lateral displacements. In this case it is possible to find the radiance distribution that, for a given index profile, minimizes the losses. We start from (38) (with $R(x) = R_0(x)$) that can be written as

$$\int_0^1 \left\{ R_0'^2(x) \int_0^{i(x)} f'^2(r) r dr - \frac{2L_B}{u^2} R_0^2(x) \right\} i^2(x) dx = 0. \quad (41)$$

Imposing a variation (with fixed extremes) in $R_0(x)$ at L_B fixed which gives a zero variation on the integral of (41), we obtain a differential equation for $R_0(x)$ which minimizes L_B

$$R_0''(x) \int_0^{i(x)} f'^2(r) r dr + \frac{i(x)}{i'(x)} R_0'(x) + \frac{2L_B}{u^2} i^2(x) R_0(x) = 0. \quad (42)$$

The value of minimum coupling loss is given by the first eigenvalue ($2L_B/u^2$) of this equation.

Solutions of (42) with $R_0(0) < \infty$ and $R_0(1) = 0$, are given for power-law profiles ($f(r) = r^q$ ($i(x) = x^{1/q}$)) by

$$R_0(x) = \frac{R_0(0)}{\sqrt{x}} \frac{\Gamma(1+q/2)}{(\Gamma_{q/2,1}/2)^{q/2}} J_{q/2}(\Gamma_{q/2,1} x^{1/q}) \quad (43)$$

which gives the minimum coupling loss

$$L_B = \frac{u^2}{4q} \Gamma_{q/2,1}^2. \quad (44)$$

In the case of parabolic profiles ($q=2$) (43) reduces to (29) (which is the steady-state distribution calculated in [7]) and the minimum coupling loss for parabolic fibers is

$$L_B \simeq 1.84 u^2. \quad (45)$$

For parabolic profiles, the coefficients of u^2 for L_A and L_B for some hypothesized radiance distributions are shown in Table 1. The first two distributions are given as inter-

Table 1: Coupling losses in parabolic index fibers with different radiance distributions

$R(x) \sim$	L_A/u^2	L_B/u^2 (with $R = R_0$)
$1 - ax^2 + (a-1)x^4$ $0 \leq a \leq 2$	$6 \cdot \frac{2-a}{4-a}$	$\frac{8}{21} \frac{23a^2 - 100a + 140}{a^2 - 7a + 16}$
$1 - x^a$ $a > 0$	$\frac{a+2}{2}$	$\frac{(a+1)(a+2)}{2a+1}$
$J_1(\Gamma_{11} \sqrt{x})/\sqrt{x}$ [7] and (29)	0.92	1.84
$J_0(\Gamma_{01} \sqrt{x}) [3]$	1.17	1.86
$Q_0(x)$ of [10]	1.10	1.93

mediate distributions between the uniform and the steady-state ones; the third seems, at present, the most realistic steady-state distribution; the last are two different distributions proposed in the literature [3, 10]. In any case, there is not a great difference among the coefficients of these three last distributions and in practice, in steady-state conditions, we have that

$$L_A \approx u^2 \quad \text{and} \quad L_B \approx 2u^2 \quad (46)$$

represent good estimates of coupling loss. This is confirmed from many measurements [9, 11, 12] in which experimental values ranging between the extremes of (46) have been found.

5 Conclusions

The problem of coupling two fibers with different core diameters, maximum numerical apertures and lateral displacement has been studied when a general radiance distribution of ray congruences is present in the fiber. Losses have been calculated both considering and not considering the additional attenuation caused in the second fiber by the joint. Moreover a method, valid for any index profile, for obtaining the actual radiance distribution among ray congruences has been proposed. Once such distribution is known, realistic numerical calculations of coupling loss have been performed. Simple analytical expressions of losses have been obtained in the particular but usual case of small error parameters.

In this case the behaviour of losses against each error parameter is (for realistic radiance distributions) quadratic. But the coupling loss which includes the additional attenuation induced in the second fiber as a function of mismatches in core radius and maximum numerical aperture results linear. Finally conditions for minimizing coupling losses have been discussed.

Anyhow, further work on this subject should be performed both experimentally (e.g. to verify the linear decay above mentioned) and theoretically (e.g. to consider radiance distribution as a function of both ray congruence variables).

References:

- [1] H. Tsuchiya, et al.: Appl. Opt. 16 (1977) 5, 1323-1331
- [2] P. Di Vita, U. Rossi: Opt. Quantum Electron. 10 (1978), 107-117
- [3] D. Gloge: Bell Syst. Tech. J. 55 (1976) 7, 905-916
- [4] S. E. Miller, A. G. Chynoweth (ed.): Optical Fiber Telecommunications, Academic Press, New York, (1979), 76-81
- [5] S. C. Mettler: Bell Syst. Tech. J. 58 (1979) 10, 2163-2182
- [6] P. Di Vita, U. Rossi: "Light propagation theory in optical fibres" in Optical Fibre Communication, CSELT (1980), Turin
- [7] R. Olshansky: Appl. Opt. 14 (1975) 4, 935-945
- [8] S. Piazzolla, G. De Marchis: Electron. Lett. 15 (1979) 22, 721-722
- [9] G. Cocito, et al.: CSELT Rapp. Tecn. 6 (1978) 2, 113-119
- [10] P. Di Vita, U. Rossi: Opt. Quantum Electron., to be published
- [11] B. Costa, et al.: Electron. Lett. 15 (1979) 18, 550-551
- [12] C. M. Miller, S. C. Mettler: Bell Syst. Tech. J. 57 (1978) 9, 3167-3180